McMaster University | Engineering Software Engineering

From Self-Play to Human Play: An Explainable Catan AI Platform

Play And Learn Catan Against The World's Strongest Catan Bot

SCAN → PLAY OUR BOT

https://rlcatan.vercel.app/

https://github.com/AlgoCatan/RLCatan

Settlers of Catan

A multiplayer strategy game where players gather resources, build, trade, and compete to reach 10 victory points.

Our Goal

Empower players of all skill levels to master Catan through *Experiential Learning* by providing a high-fidelity digital environment where they can practice, learn, and compete against sophisticated bots.

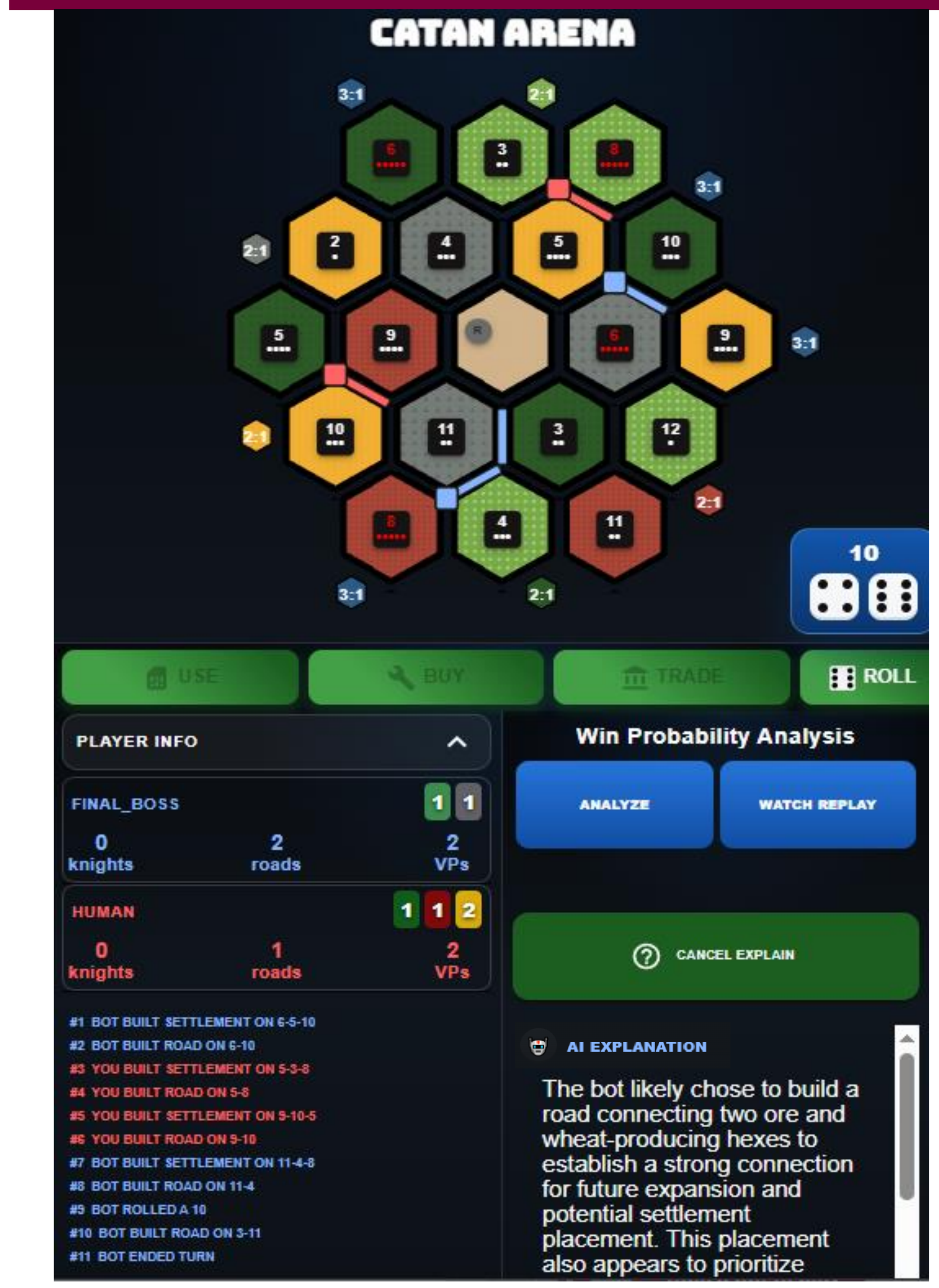
Our Motivation

Catan is the most popular modern board game, yet remains currently unsolved in the context of reinforcement learning. We hope to push the field forward with a competitive platform for Catan bots.

Plan to Achieve Goal

- Create an easy-to-use user interface.
- Train a bot to play Catan.
- Generate an explanation for every move.

Interface



PPO-Based Reinforcement Learning

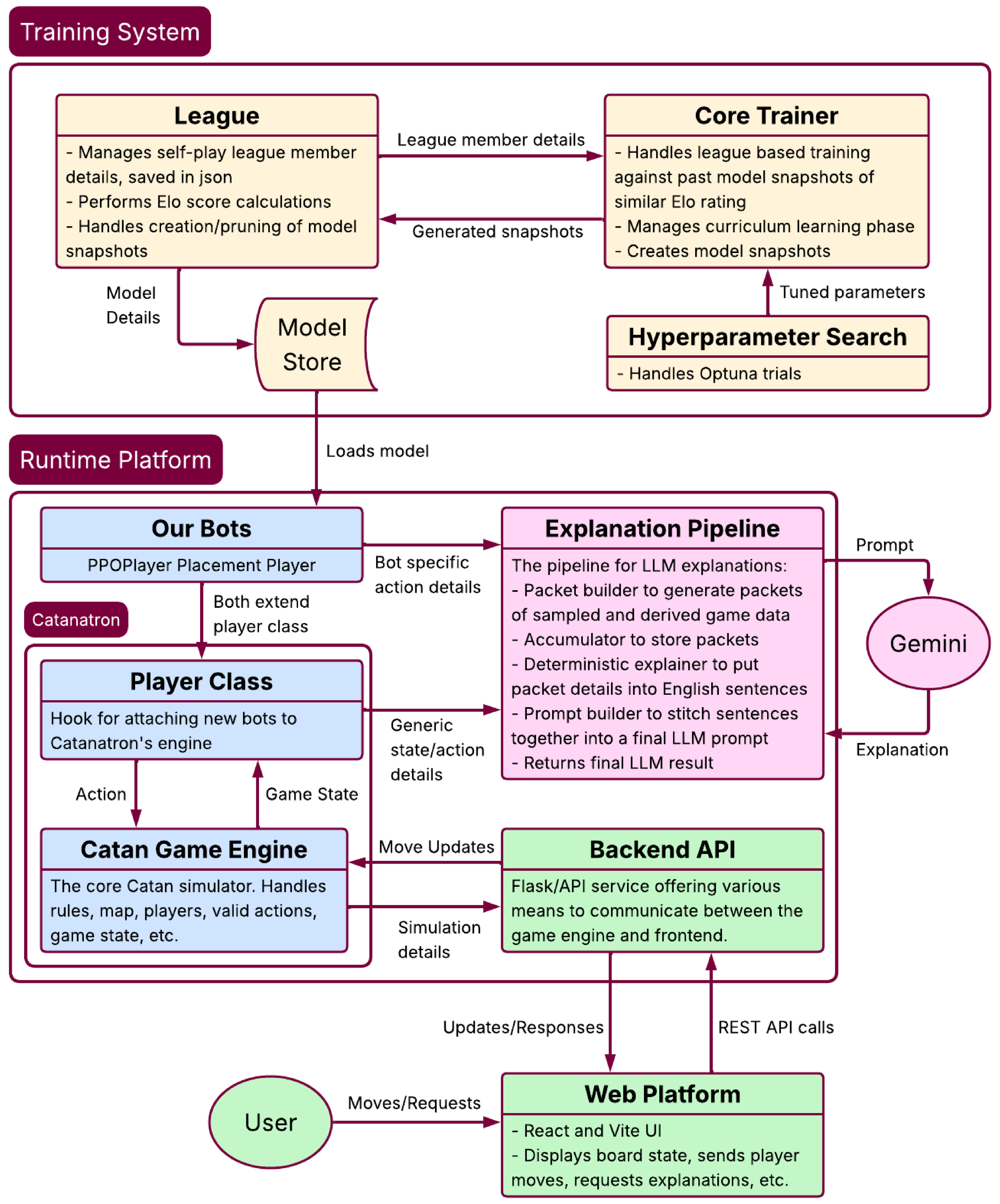
Hyper-parameter Tuning
- Engineered a parallelized Distributed Worker System on a multi-GPU server to transform hyperparameter tuning into an automated search over 10,000,000+ turns.

Reward Shaping
- Performed strategic reward shaping that provides the agent with continuous feedback, incentivizing both immediate gameplay and the long-horizon strategic milestones required to secure a victory.

Curriculum Learning
- Implemented a training pipeline that incrementally scales game complexity, allowing the agent to master foundational gameplay before advancing to the full-scale game with all features included.

Self-Play League
- Established a self-play cycle where agents continuously compete against prior versions of themselves, driving performance improvements and progressively stronger strategies over time.

Architecture



Challenges/Accomplishments

Integration of Systems

- Designing a cohesive system that connects the Catanatron simulator, reinforcement learning pipeline, and LLM explanation layer required careful architectural decisions and custom environment wrappers to ensure seamless interaction between components.

Massive State & Action Space

- Catan is an extremely large state space with over 15 octillion possible boards and 290 actions. On top of that there is stochasticity because of the randomness involved. Efficiently exploring and learning within this space is a complex task. This is why our bot took weeks to train on 1,000,000+ games.

Usability

- Making the system interpretable and useful for humans introduced additional challenges. We developed LLM-generated explanations for each agent decision, alongside readability metrics (ex. Flesch scores) to ensure clarity and accessibility.

Team AlgoCatan

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Results

Matchup 1: Final Boss vs Baseline Random Bot

	Final Boss (Our Bot)	Baseline Random Bot
WINS	100	0
Avg VP	15.09	2.68
Avg Settles	1.87	4.73
Avg Cities	0.18	4.00
Longest Road %	99%	1%
Largest Army %	7%	9%

Matchup 2: Final Boss vs Former World's Best Bot

	Final Boss (Our Bot)	Former World's Best Bot
WINS	60	40
Avg VP	12.84	11.65
Avg Settles	3.85	3.84
Avg Cities	3.55	4.00
Longest Road %	64%	36%
Largest Army %	11%	15%

Tools

python · TypeScript · HTML5 · PyTorch · OpenAI Gym · STABLE-BASELINES3 · CUDA · Optuna · Vercel · Google GenAI · React · docker